08/25
sday.

Thermodynamics

→ System.
The part of universe under observation.

→ Surrounding.
The part of universe other than system.

"System + surrounding = Universe"

→ TYPES OF SYSTEM.

Open : exchange of matter & energy.

Closed : neither energy nor matter.

Isolated : only energy.

→ PROPERTIES OF SYESTEM.

Extensive properties	Intensive properties.
depends on amnt of substance	Independent of amnt of substance.
Mass, Enthalpy, Entropy, Internal Energy (U), Volume; Heat Capacity.	Temp, Viscosity, Refractive Index, Density, specific Heat Capacity, Molar Enthalpy.

→ State function :- Functions which depends only on the final & initial value (state) and not on the path.

→ path function :- functions which depends on the path followed
eg:- Heat & Work.

⇒ Types of process :-

Isothermal process ⇒ occurs at constant temperature $\Delta T = 0$
heat change occurs. $dq \neq 0$

→ First **Adiabatic process** ⇒ no heat change, temperature changes, $dT \neq 0$

- **Isobaric process** ⇒ occurs at constant pressure. $dp = 0$
- **Isochoric process** ⇒ occurs at constant volume. $dV = 0$
- Cyclic process ⇒ process in which the system return to the original state
- Reversible process ⇒ It is process in which the direction may be reversed by a very small change in any one of the variables.
- Irreversible process ⇒ It is a process which cannot be reversed by small change in variables.

⇒ **First Law of Thermodynamics.**

* "Energy can neither be created nor destroyed although it can be converted from one form into another."

$$dE = dq - P\,dV$$

* "The total energy of the universe remains constant."

⇒ Internal energy or Intrinsic energy.

* A definite quantity of energy which depends upon factors such as chemical nature of the substance, temp and pressure.

→ Internal Energy change, ΔE.

The change in internal energy (ΔE) of a reaction is equal to heat evolved or absorbed during the rn at constant temperature and constant volume. ie, *

$$\Delta E = qV$$

Change in internal energy → Bomb Calorimeter.

⇒ **Enthalpy**

Enthalpy may be defined as the sum of the internal energy and the product of its pressure and volume.

$$H = E + PV$$

⇒ Spontaneous process.

* feasibility of process; A process which can take place by itself under a given set of condition is said to be a spontaneous process.

* The direction in which energy is lowered but also by the determined not only by the direction in which disorder is increased

⇒ **Entropy.**

* The extent of disorder in a system is expressed by a property called entropy.

* entropy change;

$$\Delta S = \frac{q_{rev}}{T}$$

⇒ **Second Law of Thermodynamics**

"The entropy of the universe always increases in the course of every spontaneous (natural) change.

"The entropy of the universe tends towards a maximum" – 2nd law

ΔS = +ve : spontaneous

ΔS = -ve : non-spontaneous

$\Delta S = 0$: equilibrium

8/9/25 Monday → **Gibb's Free Energy. (ΔG)**

* The energy available for useful work.
* For a spontaneous process, ΔG = -ve.
* " non-spontaneous process, ΔG = +ve
* For equilibrium , $\Delta G = 0$

$\Delta G = \Delta H - T\Delta S$ → Gibb's - Helmoltz eqn.

ΔH	$T\Delta S$	ΔG	process.
-ve (exo)	+ve	-ve	Spontaneous.
-ve (exo)	-ve	-ve when ΔH > $T\Delta S$ (greater)	Spontaneous
		+ve when $\Delta H < T\Delta S$ (greater)	non spontaneous
+ve	+ve	-ve	when $T\Delta S > \Delta H$ Spontaneous.

Q: $H_2O_{(s)} \longrightarrow H_2O_{(l)}$

$\Delta H_f = 6.97$ KJ/mol

$\Delta S_f = 25.4$ J/K mol at 25°C

Calculate the free energy and predict whether melting of ice is spontaneous or not at this temp.

$\Delta G = \Delta H - T\Delta S$

$= 6.97 \times 298$

T = 273 + 25

= 298 K

$= 6970 - (298 \times 25.4)$

$= 162.8$

$= -599.2$ J

$= -0.5992$ KJ

$= -0.6$ J

Since ΔG = -ve, this process is indeed spontaneous at this temp.

Q: For the r^n ; $M_2O_{(s)} \longrightarrow 2M_{(s)} + \frac{1}{2}O_{2(g)}$

ΔH = 30 KJ/mol

ΔS = 0.07 KJ/K mol

Calculate at what temp the r^n would be spontaneous.

$\Delta G = \Delta H - T\Delta S$

$\Delta H - T\Delta S < 0$

$\Delta H < T\Delta S$

$T > \frac{\Delta H}{\Delta S}$

$T > \frac{30}{0.07}$

$T > 428.5$ K

Q:- For the rⁿ; $2H_2O_{2(g)} \longrightarrow 2H_2O_{(g)} + O_{2(g)}$

$\Delta H = -211.29$ KJ/mol

$\Delta S = 131.98$ J/mol·K

Predict the feasibility of the rⁿ at 298 K.

$\Delta G = \Delta H - T\Delta S$

$\Delta G = \{-211.29 \times 1000\} - (298 \times 131.98)$

$= -211290 - 39330.04$

$= -250620.04$

$= -250$ KJ

∴ ΔG = -ve Hence rⁿ is feasible.

→ **Heat Capacity (C)**

The quantity of heat req to increase the temp of a substance by 1°C or 1K.

Substance 10/9/25 WED.

$$C = \frac{q}{\Delta t}$$

→ Specific Heat Capacity (C)

The amnt of heat req to increase the temp of a unit mass of a substance by 1K

$$C = \frac{q}{m \times \Delta t}$$

→ Molar heat Capacity (

* The amnt of heat req to ↑ the temp of 1mole of a substance by 1°C or 1K.

$$dE = dq - P \cdot dV$$ → first law.

at constant Volume ; $dV = 0$

∴ $$dE = dq$$

$$\frac{dE}{dt} = \left[\frac{dq}{dt}\right]_V$$

$$C_V = \left[\frac{dE}{dt}\right]_V$$

at constant Pressure ; $P = 0$

$$dE = dq$$

$$C_P = \left[\frac{dH}{dt}\right]_P$$

$H = E + PV$

$H = E + RT$

$dH = dE + R\,dT$

$\frac{dH}{dT} = \frac{dE}{dT} + R$

∴ $C_P = C_V + R$ i.e. $C_P - C_V = R$ Mayor's Relation.

$PV = nRT$ → 1

$PV = RT$

$$C_P/C_V = \gamma$$

⇒ Enthalpy of formation (ΔH_f)

* enthalpy change associated when one mole of a compound formed from its elements in their standard state.
* for each compound it is fixed.
* $\Delta H_f = \Delta H_{pdt} - \Delta H_{rct}$

→ * Standard Enthalpy of Formation (ΔH_f°)

$$\Delta H_f^\circ = \Delta H_{pdt}^\circ - \Delta H_{rct}^\circ$$

ΔH_f° of a r^n is same as ΔH_{pdt}° bcuz for elements in their standard state, ΔH_{rct}° is zero.

→ Enthalpy of Combustion (ΔH_c)

* Amnt of energy released when 1 mole of a substance is burned in excess oxygen.
* ΔH_c is always -ve.

eg: $C_{(g)} + O_{2(g)} \longrightarrow CO_{2(g)}$ $\quad \Delta H_c = -393.5\ KJ/mol$

$$C_6H_6 + \frac{15}{2}O_2 \longrightarrow 6CO_2 + 3H_2O$$

⇒ Enthalpy of Neutralisation

Amnt of energy associated with when one gram equivalent of an acid is neutralised with one gram equivalent of a base:

* eg: $NaOH + HCl \longrightarrow NaCl + H_2O$, $\Delta H = -57\ KJ$

$KOH + HNO_3 \longrightarrow KNO_3 + H_2O$, $\Delta H = -57\ KJ$

$NaOH + H_2SO_4 \longrightarrow Na_2SO_4 + H_2O$, $\Delta H = -57\ KJ$

★ for strong acid - strong base, $\Delta H_{neut} = -57\ KJ$.

This is bcuz complete dissociation occuring here.

eg: $Na^+ + OH^- + H^+ + Cl^- \longrightarrow Na^+ + Cl^- + H_2O$

$H^+ + OH^- \longrightarrow H_2O$

eg: $K^+ + OH^- + H^+ + NO_3^- \longrightarrow K^+ + NO_3^- + H_2O$

$H^+ + OH^- \longrightarrow H_2O$

* eg: W.A S.B
$CH_3COOH + NaOH \longrightarrow CH_3COONa + H_2O$ $\quad \Delta H = -55.5\ KJ$

Here the rest of 1.5 KJ is utilized for dissociation of weak electrolyte (CH_3COOH).

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Heat Capacity:

$Q_p = \Delta H$

$Q_v = \Delta E$

$\Delta H = \Delta E + P\Delta V$

$Q_p = Q_v + P\Delta V$

$PV = nRT$

$V = \frac{n_g RT}{P}$

$\Delta V = \frac{\Delta n_g RT}{P}$

$$Q_p = Q_v + \Delta n_g RT$$

Q:- The enthalpy of formation of methane at constant pressure at 300 K is $-75.83\ kJ$. What will be the heat of formation at constant volume. $R = 8.314\ JK^{-1}mol^{-1}$

$Q_p = -75.83\ kJ$

$T = 300K$

$R = 8.314$

$C_{(s)} + 2H_{2(g)} \longrightarrow CH_{4(g)}$

$\Delta n_g = \Delta n_{P(g)} - \Delta n_{R(g)}$

$= 1 - 2$

$= -1$

$Q_p = Q_v + \Delta n_g RT$

$Q_v = Q_p - \Delta n_g RT$

$= -75.83kJ - (-1 \times 8.314 \times 300)$

$= -75.83kJ - (8.314 \times 300)$

$= -75.83 \times 10^3 + (8.314 \times 300)$

$= -73335.8$

$= -73.33\ kJ$

Kirchoff's Relation.

→ Variation of heat of rxn with temperature.

$A \longrightarrow B$

$\Delta E = E_B - E_A$

$$\frac{d(\Delta E)}{dt} = \frac{dE_B}{dt} - \frac{dE_A}{dt}$$

$$\frac{d(\Delta E)}{dt} = C_{V_B} - C_{V_A}$$

$$\frac{d(\Delta E)}{dt} = \Delta C_V \qquad \int dx = x$$

$$d(\Delta E) = \Delta C_V \cdot dt$$

$$\int_{E_1}^{E_2} d(\Delta E) = \int_{T_1}^{T_2} \Delta C_V \cdot dt$$

$$\boxed{\Delta E_2 - \Delta E_1 = \Delta C_V (T_2 - T_1)}$$

→ Variation of Heat of rxn with temperature :-

$A \longrightarrow B$

$\Delta H = H_B - H_A$

$$\frac{d(\Delta H)}{dt} = \frac{d(H_B)}{dt} - \frac{d(H_A)}{dt}$$

$$\frac{d(\Delta H)}{dt} = C_{P_B} - C_{P_A}$$

$$\frac{d(\Delta H)}{dt} = \Delta C_P$$

$$d(\Delta H) = \Delta C_P \cdot dt$$

$$\int_{H_1}^{H_2} d(\Delta H) = \int_{T_1}^{T_2} \Delta C_P \cdot dt$$

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$$\Delta H_2 - \Delta H_1 = \Delta C_p (T_2 - T_1)$$

Q:- The enthalpy of r^n

$$N_2 + 3H_2 \longrightarrow 2NH_3 \text{ at } 300K$$

was found to be $-91.97\ KJ$. what'll be the enthalpy of at $323K$. C_p values of N_2, H_2 & NH_3 are $28.46, 28.33$ & $37.08\ JK^{-1}mol^{-1}$.

$T_1 = 300K$

$T_2 = 323K$

$\Delta H_1 = -91.97\ KJ$

$\Delta H_2 = ?$

$\Delta H_2 - \Delta H_1 = \Delta C_p (T_2 - T_1)$

$\Delta H_2 = \{\Delta C_p (T_2 - T_1)\} + \Delta H_1$

$\therefore \Delta C_p = 2 \times C_{p(NH_3)} - (C_{p(N_2)} + 3 \times C_{p(H_2)})$

$= (2 \times 37.08) - (28.46 + 3 \times 28.33)$

$= 74.16 - (113.45)$

$= -39.29$

$\Delta H_2 = \{-39.29\ (23)\} + -91.97 \times 10^3$

$= -92.87\ KJ$

9/25 Thursday. **HESS'S LAW.**

The enthalpy of r^n is same whether the r^n takes place in single step or several steps.

eg: $C + O_2 \longrightarrow CO_2$; $\Delta H = -393.5\ KJ/mol$

$C + \frac{1}{2}O_2 \longrightarrow CO$; $\Delta H = -110\ KJ/mol$

$CO + \frac{1}{2}O_2 \longrightarrow CO_2$; $\Delta H = -283.5\ KJ/mol$

Q: Calculate the heat of formation of Ethane from the following data.

i. $C + O_2 \longrightarrow CO_2$ $\Delta H = -393.5 KJ/mol$. —①

ii. $H_{2(g)} + \frac{1}{2}O_{2(g)} \longrightarrow H_2O_{(l)}$ $\Delta H = -285.8 KJ/mol$ —②

iii. $C_2H_{6(g)} + \frac{7}{2}O_{2(g)} \longrightarrow 2CO_{2(g)} + 3H_2O_{(l)}$ $\Delta H = -1560 KJ/mol$ —③

$2C_{(s)} + 3H_{2(g)} \longrightarrow C_2H_{6(g)}$; $\Delta H = ?$

$\therefore$ ①×2 + ②×3

$(-393.5) \times 2 + (-285.8) \times 3$

$\Downarrow$

$-787 + -857.4$

$\Downarrow$

-1644.4

$2CO_2 + 3H_2O \longrightarrow 7/2 O_2 + C_2H_6$

$\therefore -2\times\Delta H_1 + 3\Delta H_2$

$2C + 2O_2 \longrightarrow 2CO_2$ +

$3H_2 + 3/2 O_2 \longrightarrow 3H_2O$

$2C + 2O_2 + 3H_2 + 3/2 O_2 \longrightarrow 2CO_2 + 3H_2O$ – ④

$2CO_2 + 3H_2O \longrightarrow C_2H_6 + 7/2 O_2$

$2C + 3H_2 \longrightarrow C_2H_6$

$\therefore \Delta H = -84.4$ KJ/mol

Q: Calculate the enthalpy of formation of Benzene from the following data.

i) Standard enthalpy of Combustion of Benzene $\Delta H_c^\circ = -3266$ KJ

ii) Standard enthalpies of formatn of CO_2 & H_2O are -393.5 & -286 KJ respt.

$6C + 3H_2 \longrightarrow C_6H_6 \quad \Delta H_f = ?$

① — $C_6H_6 + \frac{15}{2} O_2 \longrightarrow 6CO_2 + 3H_2O \quad \Delta H_c^\circ = -3266$ KJ

② — $C + O_2 \longrightarrow CO_2 \quad \Delta H = -393.5$ KJ

③ — $H_2 + 1/2 O_2 \longrightarrow H_2O \quad \Delta H = -286$ KJ

② × 6 + ③ × 3 $\qquad -2361 + -858$

$$6C + 6O_2 \longrightarrow 6CO_2$$
$$3H_2 + \frac{3}{2}O_2 \longrightarrow 3H_2O +$$

$$6C + \frac{15}{2}O_2 + 3H_2 \longrightarrow 6CO_2 + 3H_2O \quad \text{—④} \quad \Delta H = -3219$$

$$\therefore \cancel{6C} + \frac{15}{2}\cancel{O_2} + 3H_2 \longrightarrow \cancel{6CO_2} + \cancel{3H_2O} -$$
$$\cancel{6CO_2} + \cancel{3H_2O} \longrightarrow C_6H_6 + \frac{15}{2}\cancel{O_2}$$

$$6C + 3H_2 \longrightarrow C_6H_6$$

$$\Delta H = -3219 + 3266$$

$$\therefore \quad = 47\ KJ/mol$$

Q: Calculate the enthalpy of formation of acetic acid (CH_3COOH). if its enthalpy of combustion is 867 KJ/mol. The enthalpy of format^n of CO_2 & H_2O -393.5 & -285.8 KJ/mol resp.

$$2C + 2H_2 + O_2 \longrightarrow CH_3COOH \qquad \Delta H_f = ? \quad \cancel{\Delta H_f = 867\ KJ/mol}$$

① — $CH_3COOH + 2O_2 \longrightarrow 2CO_2 + 2H_2O \quad \Delta H = -867$

② — $C + O_2 \longrightarrow CO_2 \quad \Delta H = -393.5$

③ — $H_2 + \frac{1}{2}O_2 \longrightarrow H_2O \quad \Delta H = -285.8$

② × 2 + ③ × 2

$$2C + 2O_2 \longrightarrow 2CO_2 +$$
$$2H_2 + O_2 \longrightarrow 2H_2O$$
$$2C + 2H_2 + 3O_2 \longrightarrow 2CO_2 + 2H_2O \qquad \Delta H = -1358.6$$

$$2C + 3H_2 + 2O_2 \longrightarrow 2CO_2 + 2H_2O$$

$$2CO_2 + 2H_2O \longrightarrow CH_3COOH + 2O_2$$

$$2C + 3H_2 + O_2 \longrightarrow CH_3COOH$$

$$\Delta H_f = -491.6\ KJ/mol$$